

A 45 nm CMOS-SOI Monolithic Photonics Platform With Bit-Statistics-Based Resonant Microring Thermal Tuning

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Abstract—The microring resonator is critical for dense wavelength division multiplexed (DWDM) chip-to-chip optical I/O, enabling modulation and channel selection at the μm -scale suitable for a VLSI chip. Microring-based links, however, require active tuning to counteract process and thermo-optic variations. Here, we present a bit-statistical tuner that decouples tracking of optical one- and zero-levels to realize non-dc-balanced data transmission, an “eye-max”-locking controller, and self-heating cancellation without need for a high-speed sensing frontend. We implement the tuner on a 45 nm CMOS-SOI process with monolithically integrated photonic devices and circuits. The tuner consumes 0.74 mW in the logic while achieving a record 524 GHz (> 50 K temperature) tuning range at 3.8 $\mu\text{W}/\text{GHz}$ heater efficiency. To our knowledge, this is the highest range and heater efficiency reported by an on-chip closed-loop thermal tuner to date. The tuner integrates with a 5 Gb/s 30 fJ/bit monolithic microring transmitter, achieving wavelength-lock and immunity to both tracking failures and self-heating events caused by arbitrary, non-dc-balanced bitstreams. In addition, the tuner provides critical functionality for an 11- λ DWDM transmitter macro capable of 11×8 Gb/s bandwidth on a fiber. Together with the transmitter, a 10 Gb/s on-chip monolithic optical receiver with 10^{-12} BER sensitivity of 9 μA at 10 Gb/s enables a sub-pJ/bit 5 Gb/s optical chip-to-chip link, with the bit-statistical tuner providing thermally robust microring operation.

Index Terms—Microring resonator, monolithic, receiver, silicon photonics, transceiver, thermal tuning.

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I. INTRODUCTION

SILICON photonics-enabled optical interconnect can overcome the traditional bandwidth- and power-density tradeoffs faced by high-speed chip-to-chip electrical links [1]. The ability to integrate with VLSI electronics, either monolithically on the same chip [2]–[4] or heterogeneously on separate chips via three-dimensional (3-D) stacking [5], [6] or system-in-package integration [7], [8], allows silicon photonics to provide a scalable interconnect for future memory systems [9], [10], [42], high-radix network switch chips [11], and core-to-core crossbars [12], [13].

Key to the realization of bandwidth densities and energy efficiencies promised by silicon photonics is the silicon microring resonator. Microrings act as μm -scale optical notch filters, rejecting wavelengths that resonate in the ring while passing wavelengths that do not. Voltages applied across microrings doped with a p - n or p - i - n junction modulate the ring's free carrier concentration and shift the microring's stop band via the free carrier plasma dispersion effect [14], [15]. This can be used to ON-OFF key (OOK) a carrier wavelength close to resonance, creating a compact, CMOS-compatible [15], and channel-selective electro-optic modulator. In a receiver, a microring performs channel selection for a broadband photodetector (PD) or acts directly as a resonant PD [4], [16]. These properties allow microrings to enable dense wavelength division multiplexing (DWDM), where multiple streams of data are carried by closely spaced carrier wavelengths sharing a single waveguide or fiber (Fig. 1), on silicon chips.

Despite all these advantages, microring-based DWDM has yet to be widely adopted due to inadequate progress on the mitigation of thermal and process variations on a microring's resonant wavelength (λ_0). A typical silicon microring with $Q = 10,000$ and a nominal λ_0 in the 1180–1550 nm (254–193 THz) band suffers > 3 dB of signal loss for a λ_0 offset of just 0.1 nm. Silicon microrings experience 0.15–1.8 nm [17] of ring-to-ring λ_0 variations from process variations and 0.05–0.1 nm/K [18], [19] of thermo-optic λ_0 shifts (-10 GHz/K [20]), indicating that a microring-based link cannot tolerate even a 1 K change in temperature without active λ_0 stabilization. Fortunately, the strong thermo-optic effect enables process variations to be compensated through a temperature change of a few Kelvin, popularizing thermal tuning of λ_0 through microring-embedded resistive heaters [8], [18], [21]–[26].

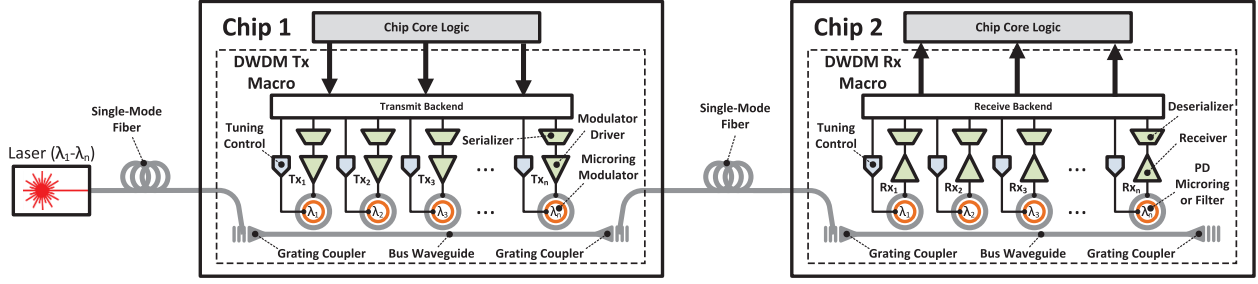


Fig. 1. Chip-to-chip DWDM link using microring modulators and resonant microring detectors. An external multiwavelength laser source supplies all wavelengths used by the link. In addition to transmitter and receiver circuitry, microrings require active tuning control.

Thermal tuning, however, typically comes at a high static power cost of 5–10 mW/nm [4], [8], [27], [28]. Moreover, efforts to make microring heater tuning more power efficient—through thermal isolation techniques such as substrate removal [7], [23], [26], [29] or chip-to-chip spacers [7]—are hampered by the exacerbated impact of ring self-heating from the laser, which causes microring optical bistability [24], [30], [31] and transient eye closure under modulation [32]. Challenges also remain for the closed-loop temperature controller. Due to the OOK nature of the data, wavelength-locking controllers that rely on averaged drop-port (or thru-port) power monitoring [25], [28], [33] cannot distinguish between a change in power due to a change in the ratios of *zeros-to-ones* in the datastream from a change in power due to a drift in λ_0 . Tuning controllers with a high-bandwidth power sensor [4], [27], [34] can decouple these two effects with appropriately added logic, but are less sensitive and unattractive for transmit-side tuners where a high-bandwidth receive frontend is not available for free.

In this paper, we present a new thermal tuner architecture that adopts bit statistics from the data stream to perform decoupled optical *one-level* and *zero-level* tracking with a low-bandwidth analog frontend. In addition to behaving appropriately for arbitrarily non-dc-balanced data patterns, the additional level-tracking information enables an “eye-max” controller design that automatically finds and locks a modulator ring to the point providing the greatest transmit eye height and a self-heating cancellation module that mitigates the impact of transient self-heating effects. We demonstrate our concept in a monolithic photonics platform fabricated in a commercial 45 nm CMOS-SOI process [23], taking advantage of the tight circuit-photonic integration to implement the pieces of the controller in a full transceiver. We leverage the platform’s postprocess substrate removal to achieve high tuning efficiency and tunable range, while using the controller to eliminate the self-heating effects that accompany the high tuning efficiency. Finally, the integration of thermal tuner on the platform enables an auto-locking 11 λ DWDM transmitter and a chip-to-chip optical link demonstration, which, to our knowledge, are the first such demonstrations in a native commercial CMOS process.

This paper is organized as follows. Section II highlights key thermal properties of microring resonators. Section III presents the proposed bit-statistical thermal tuner. Section IV introduces the 45 nm SOI CMOS photonic platform and outlines the tuner implementation. Section V describes the experimental results

and Section VI compares with prior art. Section VII concludes this paper.

II. THERMAL PROPERTIES OF MICRORING RESONATORS

We can fit the optical thru-port transmission ($\alpha(\lambda)$) of a ring resonator around a single resonance with wavelength λ_0 through a Lorentzian line shape [2], [32]

$$\alpha(\lambda) = 1 - \gamma(\lambda) = 1 - \frac{A}{1 + 4 \left(\frac{\lambda - \lambda_0}{\Delta\lambda} \right)^2} \quad (1)$$

where λ_0 is the ring’s resonant wavelength and $\Delta\lambda$ is the linewidth of the resonance. The ring’s Q -factor is equivalent to $\lambda_0/\Delta\lambda$. The constant A is indicative of device’s intrinsic extinction. $A = 1$ when the device is perfectly critically coupled [4] such that $\alpha(\lambda_0) = 0$. Note the definition of the quantity $\gamma(\lambda)$, which is the fraction of input power “lost” by the ring as a function of the laser wavelength (λ). This can include radiative losses, absorption by the ring, or coupling to a drop port.

As microring resonators absorb a fraction of the input laser power, they experience self-heating power from laser, which we quantify as

$$P_S = \kappa \cdot P_L \cdot \gamma(\lambda) \quad (2)$$

where P_L is the input port laser power and κ indicates the fraction of $P_L \cdot \gamma$ that is absorbed and converted to heat (typical κ values vary from 0.3 to 0.9 [24]). P_S will heat up the ring and change λ_0 through the microring heating efficiency, H_T (units of nm/mW). Substituting $\gamma(\lambda)$ using (1), we arrive at an expression for λ_0 which incorporates self-heating effects

$$\lambda_0 = \lambda'_0 + H_T \cdot P_S \quad (3a)$$

$$= \lambda'_0 + H_T \cdot \kappa \cdot P_L \cdot \frac{A}{1 + 4 \left(\frac{\lambda - \lambda_0}{\Delta\lambda} \right)^2} \quad (3b)$$

where λ'_0 is the nominal resonance of the ring, defined as λ_0 when $P_S = 0$ (no self-heating).

To understand the behavior of (3), we can adopt small-signal analysis to linearize γ with respect to changes in λ_0 and account for any other perturbations to the system through the quantity $d\lambda'_0$

$$d\lambda_0 = d\lambda'_0 + H_T \cdot \kappa \cdot P_L \cdot \frac{\partial \gamma}{\partial \lambda_0} \cdot d\lambda_0. \quad (4)$$

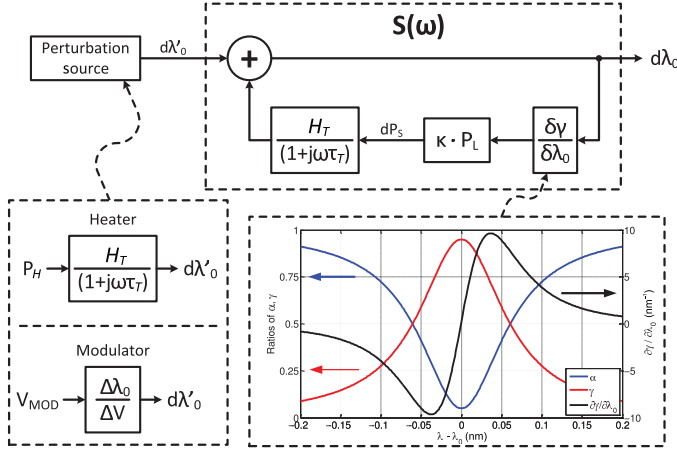


Fig. 2. LTI system model showing the feedback relationship of laser self-heating. Effects from modulation, heaters, or external temperature changes impact the microring resonance through $d\lambda'_0$.

Equation (4) is identical to the expression for an LTI system with first-order feedback (Fig. 2), allowing analysis of microring self-heating as a feedback system. Intuitively, feedback is inherent to the system, as a change in λ_0 causes a change in P_S that again changes λ_0 . The system has a transfer function $S(\omega)$ with a DC loop gain given by

$$G_L = H_T \cdot \kappa \cdot P_L \cdot \frac{\partial\gamma}{\partial\lambda_0}. \quad (5)$$

As κ , P_L , and H_T are positive physical terms, $\partial\gamma/\partial\lambda_0$ alone determines whether self-heating contributes positive ($G_L > 0$) or negative ($G_L < 0$) feedback. When the ring is *red-biased* (laser $\lambda < \lambda_0$), $\partial\gamma/\partial\lambda_0 < 0$, implying negative feedback. When *blue-biased* (laser $\lambda > \lambda_0$), $\partial\gamma/\partial\lambda_0 > 0$, implying positive feedback. The transfer function can become unstable ($G_L > 1$) when *blue-biased* under sufficiently high P_L . When unstable, any perturbation to the ring is continuously amplified by self-heating and the ring resonance drifts away from the original point until $\partial\gamma/\partial\lambda_0$ drops enough such that $G_L < 1$. This leads to thermo-optical bistability; given constant P_L , a ring biased in the bistable regime will settle to one of two thermally stable steady states.

Fig. 3 shows self-heating stability curves for different P_L , with each curve marking the combinations of (λ, λ_0) that satisfy (3). Bistable regimes, whereby multiple λ_0 solutions exist for a single laser λ , can be observed. In the case where $P_L = 1$ mW, the ring is bistable over a large range of laser λ . Within the bistable regime, the ring can settle into either state B or state C (where $G_L < 1$). The (λ, λ_0) combinations with $G_L > 1$ form the metastable state. Here, any nonzero perturbation will push the microring into state B or C due to the positive G_L . To reach the bistable state B, the system must first be *red-biased* into state A. λ must then increase to move into state B. Similarly, to reach state C, the ring must first be *blue-biased* into state D followed by a decrease in λ to move into state C. The range of λ with bistable behavior shrinks with smaller P_L and disappears at $P_L = 0.2$ mW, though self-heating still creates noticeable curve asymmetry.

Bistability creates hysteresis in how λ_0 responds to an embedded microring heater used for tuning (Fig. 4). If the heater power is set by a digital-to-analog converter (DAC), we obtain either enhancement of the DAC resolution (in nm/LSB or GHz/LSB) for a *red-biased* ring or degradation of the DAC resolution for a *blue-biased* ring near the resonance. A DAC driving a *blue-biased* ring (assuming it is stable) needs extra bits of precision, whereas a DAC driving a *red-biased* ring can be made coarser as it gains extra precision from the self-heating for free.

Given that temperature change from heating is not instantaneous, Fig. 2 models the transient heating behavior as a first-order low-pass filter with a thermal time constant τ_T . As the low-pass thermal time constant τ_T appears in the loop gain, $S(\omega)$ will have a zero at $1/\tau_T$ and a pole at $(1 - G_L)/\tau_T$. Under *red-bias* (negative feedback), the self-heating relationship attenuates low frequencies at a cutoff frequency that grows with $1 - G_L$. Under *blue-bias* (positive feedback), self-heating provides low-frequency gain. Treating modulation as a perturbation of $d\lambda'_0$, the negative feedback from self-heating for a *red-biased* ring creates eye closure akin to baseline wander noise in ac-coupled systems with a high low-frequency cutoff. For a *blue-biased* ring, the added low-frequency gain adds ISI. Although these self-heating effects may be tolerable for encoded, dc-balanced data transmission, arbitrary link traffic that is not dc-balanced will suffer from unacceptable eye closure if self-heating is left unchecked.

III. BIT-STATISTICAL THERMAL TUNER

In this section, we propose a thermal tuner that can correctly handle non-dc-balanced data by extracting the optical power levels of *ones* and *zeros* independently without the use of high-bandwidth receiver-like frontends, which are costly in power and limited in sensitivity.

A. Bit-Statistical Tracking Method

Consider the case of a photodetector (PD) placed on the drop-port of microring transmitting or receiving NRZ data, connected to a photocurrent-to-voltage capacitive integrator. The voltage output of the integrator after an interval of N bits with N_0 *zeros* and N_1 *ones* is

$$v_N = \frac{t_{\text{bit}}}{C} (N_0 \cdot i_0 + N_1 \cdot i_1) \quad (6)$$

where t_{bit} is the bit-time and the values of i_0 and i_1 are photocurrent levels corresponding to *zeros* and *ones*, respectively. i_0 and i_1 are approximately constant over the interval if $N \cdot t_{\text{bit}} \ll \tau_t$. If we integrate two N -bit intervals, at different times t_a and t_b , but with $t_a - t_b \ll \tau_T$ such that $i_{0,1}(t_a) \approx i_{0,1}(t_b)$, then

$$\begin{aligned} v_N(t_a) &= \frac{t_{\text{bit}}}{C} [N_0(t_a) \cdot i_0 + N_1(t_a) \cdot i_1], \\ v_N(t_b) &= \frac{t_{\text{bit}}}{C} [N_0(t_b) \cdot i_0 + N_1(t_b) \cdot i_1]. \end{aligned} \quad (7)$$

If $N_{0,1}(t_a) \neq N_{0,1}(t_b)$, the two expressions are linearly independent and we can solve for i_0, i_1 .

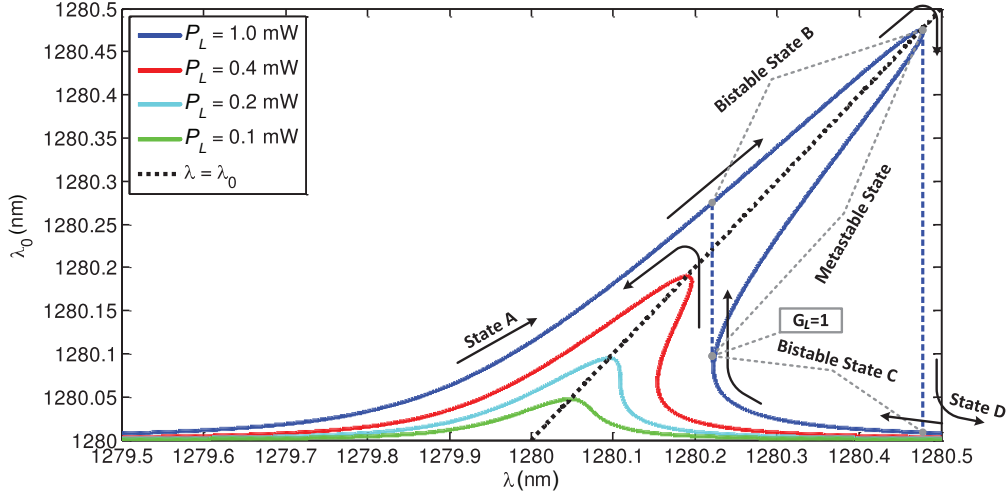


Fig. 3. Microring stability curves for a ring in steady state. A plot of λ_0 versus the laser wavelength (λ) at different laser powers (P_L). The line where $\lambda = \lambda_0$ is also shown. The ring has $\lambda'_0 = 1280$ nm, $Q = 10,000$, $H_T = 1$ nm/mW, and $\kappa = 0.5$. We annotate the region of bistability for the case of 1 mW of laser power, showing the range of wavelengths with two stable states and the metastable state. The directions of the arrows show that the trajectory λ_0 takes if we sweep the laser from low λ to high λ and from high λ to low λ .

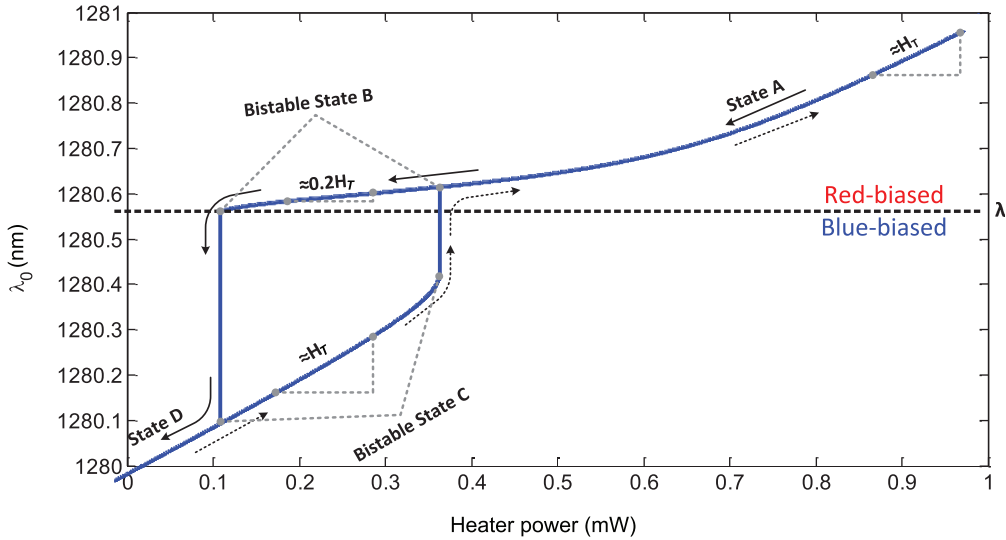


Fig. 4. Simulated heater tuning curve showing hysteresis behavior and nonlinear response of λ_0 to heater power. The dashed arrow lines indicate the path taken if the heater is swept from low power to high power, the solid arrow lines indicate the path if swept from high power to low power. When the heater is OFF, the ring's $\lambda_0 = 1280$ nm and the laser is parked at $\lambda = 1280.6$ nm. Depending on the output heater power and the bistable state, the ring may be either *red-biased* ($\lambda_0 > 1280.6$ nm) or *blue-biased* ($\lambda_0 < 1280.6$ nm). $P_L = 1$ mW, $H_T = 1$ nm/mW, and $\kappa = 0.5$.

We digitize the integrated voltage v_N through an A-to-D conversion with ratio of G_A . Variables L_0 and L_1 are digital representations of the *zero*-level and *one*-level, respectively:

$$\begin{aligned} L_N &= v_N \cdot G_A, \quad L_0 = \frac{G_A \cdot N \cdot t_{\text{bit}}}{C} \cdot i_0, \\ L_1 &= \frac{G_A \cdot N \cdot t_{\text{bit}}}{C} \cdot i_1. \end{aligned} \quad (8)$$

Equation (7) can then be rewritten as

$$\begin{aligned} L_N(t_a) &= \frac{1}{N} [N_0(t_a) \cdot L_0 + N_1(t_a) \cdot L_1], \\ L_N(t_b) &= \frac{1}{N} [N_0(t_b) \cdot L_0 + N_1(t_b) \cdot L_1]. \end{aligned} \quad (9)$$

We can reformulate L_1 and L_0 through a *level difference* term, $L_d = L_1 - L_0$

$$\begin{aligned} L_d &= \frac{N [L_N(t_a) - L_N(t_b)]}{N_0(t_b) - N_0(t_a)}, \quad L_1 = L_N + \frac{N_0}{N} \cdot L_d, \\ L_0 &= L_1 - L_d. \end{aligned} \quad (10)$$

The calculation of L_1 and L_0 through L_d decouples updates to L_d from updates to L_1 and L_0 ; although L_d cannot always be computed [division by zero when $N_0(t_a) = N_0(t_b)$], L_1 and L_0 can always update using the current value of L_d . L_1 and L_0 calculated without an up-to-date L_d will still have the same local monotonicity to enable locking to the correct point, as explained next.

Consider the case of a transmit ring wavelength-locked by a *lock-to-reference* controller, which keeps the ring at a point where $L_1 = L_{\text{ref}}$. For thermal stability, the controller keeps the ring *red-biased*, red-shifting λ_0 if $L_1 > L_{\text{ref}}$, and blue-shifting

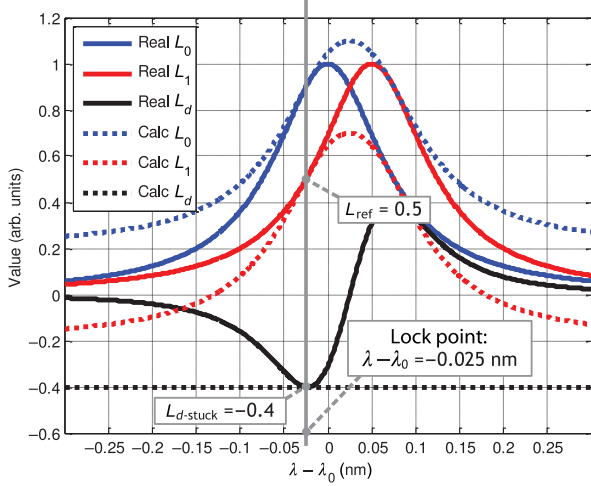


Fig. 5. Real L_0 , L_1 , and L_d versus calculated L_0 , L_1 , and L_d , plotted at various laser wavelengths (λ) relative to the nominal resonance λ_0 . The calculated values assume that L_d cannot update and is stuck at the value of the original lock point $L_{d\text{-ref}}$. The desired lock point is indicated by the vertical line, corresponding to a red-biased ring with $\lambda - \lambda_0 = 0.025$ nm. Plots are for a 50% ratio of zeros in the data sequence.

if $L_1 < L_{\text{ref}}$. At the lock point, $L_d = L_{d\text{-ref}}$. Suppose now that the transmitted data changes to a clock waveform ($N_0(t_a) = N_0(t_b)$), such that L_d cannot update and is stuck at $L_{d\text{-ref}}$. Fig. 5 shows the real versus the calculated (where L_d is stuck) values of L_0 , L_1 , and L_d . Notice that if the ring drifts off the lock point ($\lambda - \lambda_0 = -0.025$ nm), both the real and the calculated L_1 change in the same direction to be greater than or equal to L_{ref} . The lock-to-reference controller will thus make the same decision with either the real or the calculated L_1 , returning the ring back to lock point, where both the real and the calculated $L_1 = L_{\text{ref}}$. This holds true unless the resonance drifts too far away from the lock point ($\lambda - \lambda_0 > 0.025$ nm), which a locked controller will never let happen. Although the example assumes a clock waveform, this property holds true for arbitrary patterns.

This approach achieves pattern independence; by decoupling L_d calculations, the tracker always provides sufficient information for the controller to lock the ring, regardless of the transmitted or received bit pattern. Second, updates to L_d are not necessary in locked operation; once the tracker learns L_d at the lock point (during initialization), only the circuits responsible for calculating L_0 and L_1 (based on a fixed L_d) need to remain active to maintain lock.

B. Eye-Max Lock Controller

The controller sets the power output of the microring heater P_H to wavelength-lock the ring to a lock point. Suppose L_{opt} is the value the controller optimizes for (maximize or minimize) during lock point selection and L_{track} is the value the controller tracks to maintain lock. Decoupled L_{track} and L_{opt} allows selection of an L_{track} that is monotonic where L_{opt} is optimal. With a bit-statistical tracker, L_{opt} and L_{track} can each be selected to be L_0 , L_1 , or L_d . Since L_d is indicative of the transmit eye height, the controller picks the “eye-max” lock point if we set $L_{\text{opt}} = L_d$. From Fig. 5, L_0 and L_1 are both monotonic around optimal L_d and are suitable for L_{track} .

Fig. 6 illustrates the simulated state of the tuning controller as it transitions through the lock process. On power-up, λ_0 is (usually) at a large wavelength offset to the laser wavelength λ , such that the ring is completely OFF-resonance. *Init* switches the transmitted data to a training sequence favorable to the operation of the bit-statistical tracker (L_d updates frequently) and sets P_H to an initial value $P_{H\text{-init}}$. The controller then enters the *search* state which steps P_H in large strides until L_{track} is above a set threshold R_{sweep} , indicating that λ_0 is close to λ . Next, *sweep* steps P_H by small steps as the controller maps the shape of the resonance, finding $P_{H\text{-opt}}$, $L_{\text{track-opt}}$, and $L_{\text{opt-opt}}$ at the optimal lock point, where L_{opt} is maximized or minimized. When L_{track} is again smaller than R_{sweep} , the controller exits the *sweep* state. The *reset* and *return* states set $P_H = P_{H\text{-init}}$ then again to $P_H = P_{H\text{-opt}}$, ensuring return to the optimal lock point found under optical bistability. The *lock* state maintains the lock under thermal perturbations. Once in *lock*, the controller tells the tracker to freeze L_d and arbitrary data can be sent.

C. Transient Self-Heating Cancellation

For a modulator ring transmitting OOK data, if r_0 is the ratio of zeros in the transmitted data (averaged over an interval), then the amount of data-dependent laser self-heating power is

$$\Delta \overline{P_S} = \kappa \cdot P_L \cdot r_0 \cdot [\gamma_0(\lambda) - \gamma_1(\lambda)]. \quad (11)$$

Large self-heating power perturbations occur whenever r_0 changes, affecting the ring through τ_T and far faster than ambient temperature variations. To prevent transient eye closure due to the sudden power change, we can use the heaters to deliver an amount of power, P_C , exactly opposite in sign to $\Delta \overline{P_S}$ to negate it. This power would be added to the power output specified by the controller and changes quickly based on r_0 . In order to measure r_0 , we can leverage bit-statistics from the datapath, so that $r_0 = N_0/N$ for an appropriate choice of N (which sets the bandwidth of the cancellation loop). As L_d is indicative of the difference in the amount of power resonating in the ring between the *zero* and *one* states, it is intuitively proportional to $P_L \cdot (\gamma_1 - \gamma_0)$ and hence L_D

$$P_C = \kappa \cdot \frac{P_L}{N} \cdot [\gamma_1(\lambda) - \gamma_0(\lambda)] \cdot N_0 = K_C \cdot L_d \cdot N_0. \quad (12)$$

Note that L_D and N_0 fully capture the dynamic behavior of P_C . K_C is a proportionality constant that does not change with time, lock position, or laser power. Equation (12) also holds true for a receive-side filter ring, though self-heating cancellation may not be necessary there, since receive filters sit downstream from the modulators and resonate far less power.

IV. MONOLITHICALLY INTEGRATED 45 NM SOI PLATFORM INTEGRATION

We implement our thermal tuner on a commercial 45 nm CMOS-SOI process. All photonic devices are monolithically integrated and the designs conform to an existing (purely electrical) design flow [23], with no changes made to the native

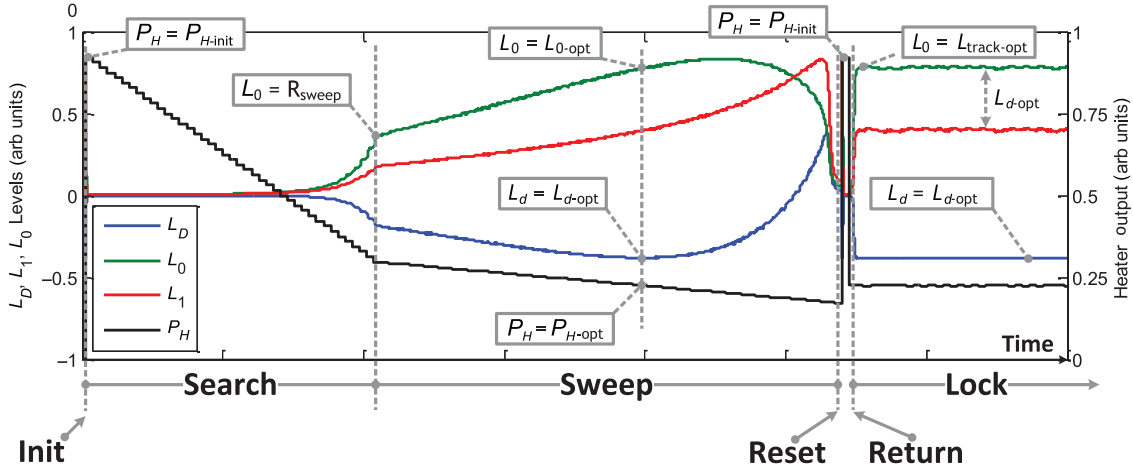


Fig. 6. Simulated controller state during the lock-on process, showing the L_0 , L_1 , and L_D (uses left axes) levels as well the heater output P_H (uses right axes) as the controller transitions from *init* to *lock*. The example uses $L_{opt} = L_D$ and $L_{track} = L_0$, hence $L_{opt-opt} = L_{d-opt}$ and $L_{track-opt} = L_{0-opt}$.

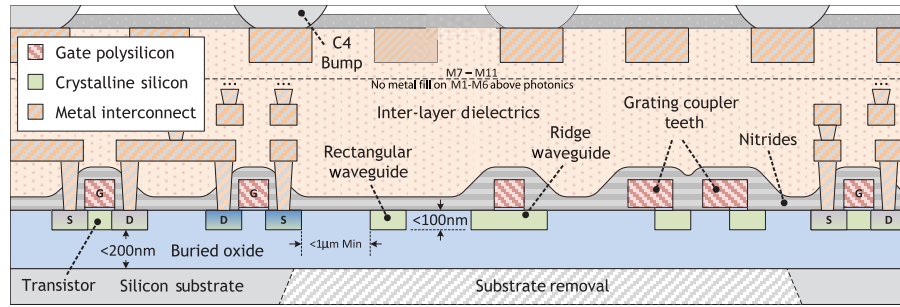


Fig. 7. Cross section of the 45 nm SOI platform. Transistors and optical waveguides are formed by patterning the c-Si layer, with polysilicon also available to break vertical symmetry. A thin-BOX layer separates the c-Si layer from the chip substrate. Metal fill blockages on M1–M6 are added to above photonics cells to prevent metals too close to waveguides. Figure is not drawn to scale.

process (Fig. 7). The process frontend consists of a crystalline silicon (c-Si) layer, patterned to form the body of transistors and high-index optical waveguide cores. The crystalline silicon layer is sandwiched between the thin buried oxide (BOX) on the bottom and, on top, layers of interlayer dielectrics and silicon-nitrides used for strain engineering. As the BOX layer is not sufficiently thick to isolate the c-Si waveguide core from the silicon substrate, we postprocess the chips after flip-chip die-attach to remove the substrate [23], [29]. Transistors are unaffected by substrate removal [23] and all existing foundry IP, timing libraries, and simulation models remain valid. The minimum separation between transistors and optical devices is sub-1 μm , limited only by the distance where the waveguide optical mode begins to evanescently couple into the transistor silicon.

The transmitter (Fig. 8) integrates both transmit datapath components and the bit-statistical thermal tuner. The modulator is a “spoked-ring” microring [26], [35]. To enable closed-loop thermal tuning, we embed a 500 Ω resistive heater and terminate the weakly coupled drop-port with a PD connected to the thermal tuner frontend. An 8-to-1 mux-latch-tree serializer, synthesized using CMOS logic standard cells, serializes the outputs from the data generators (2^{31} PRBS or 64-bit pattern) and provides a divide-by-4 clock to the rest of the transmitter. A chain of four standard cell logic inverters (drive buffers) drive the spoked-ring modulator single-endedly from the

cathode (*n*-doped) terminal. The anode (*p*-doped) terminal is connected to a bias voltage V_{BIAS} . The driver buffers provide a swing across the device equivalent to the drive buffer supply voltage V_{DD} .

The analog frontend in the tuner integrates photocurrent from the modulator’s drop-port PD onto a capacitor over an interval of 16–128 bit times. Both the integration interval and integration capacitance are knobs for tuning the integrator’s dynamic range. An SAR ADC converts the output voltage to a 6-bit digital value. The frontend controller coordinates the *reset*, *integrate*, and *evaluate* actions of the integrator and ADC. The synthesized digital backend implements the tuning controller and arithmetic calculations. The N_0 counter taps from the transmit datapath and outputs two counts: 1) N_0 , the number of *zeros* that were in the 16–128 bit integration interval, and 2) all_N_0 , the number of *zeros* every 512 bits. The aligner times the start of the tuning frontend *integrate* period with the reset of the N_0 counter, and the end of *integrate* with the load of the N_0 flip-flop (accounting for the delay through the transmit path to the drop-port PD), such that N_0 counts only the bits that affected the integrated interval. The tracker calculates L_0 , L_1 , L_d through (10). To avoid a digital divider and to provide noise filtering, the calculation of L_d is performed via a bang-bang loop, updated once every 512 bits (256 cycles). When *Lock* is asserted (by the controller), the tracker disables the loop to hold L_d constant. Calculations for L_1 and L_0 use the sums of

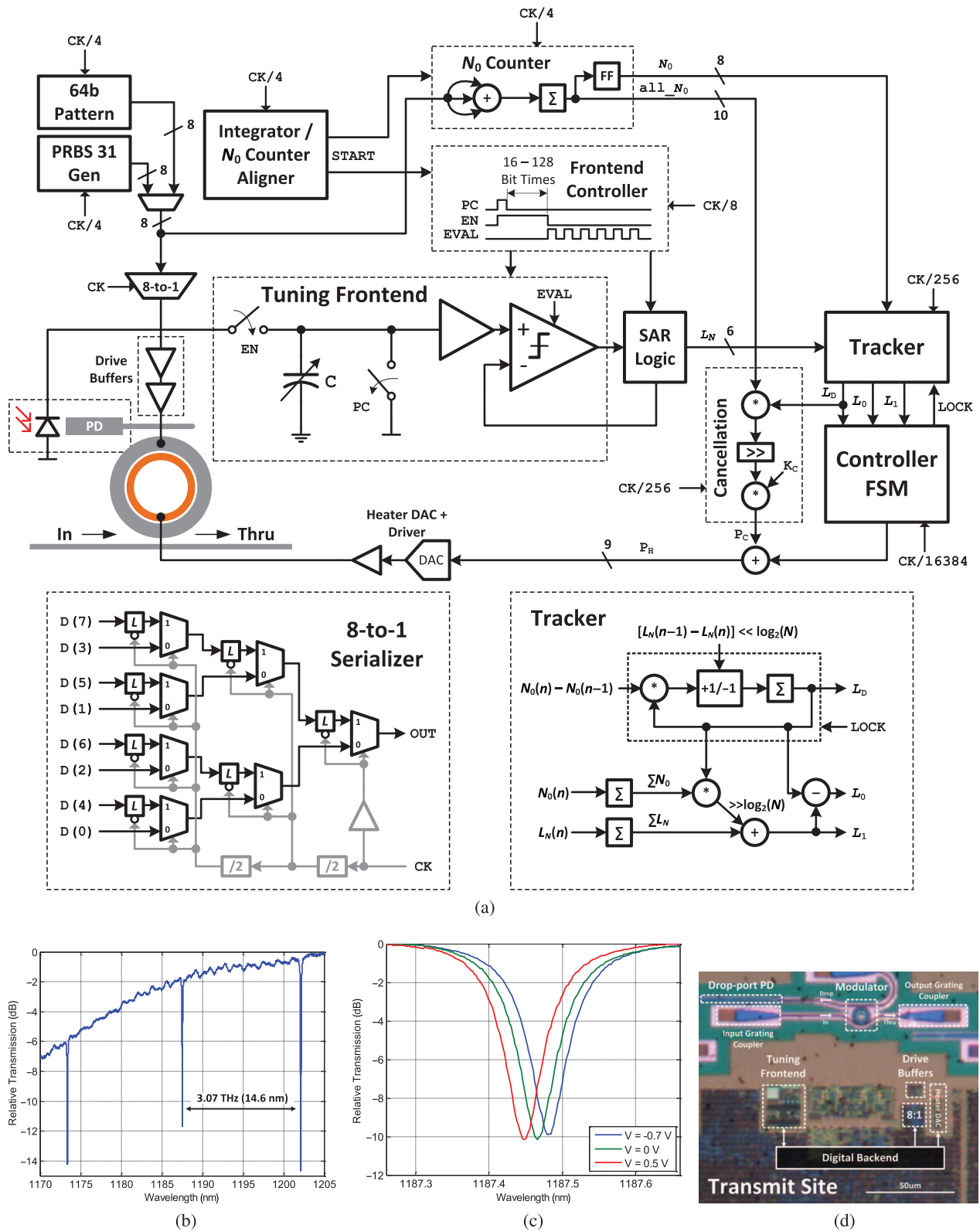


Fig. 8. (a) Schematic of the transmitter with a bit-statistical thermal tuner with insets showing the implementations of the 8-to-1 transmit serializer and the calculations performed in the bit-statistical solver. (b) Optical through-port transmission spectra showing the free spectral range of the spoked ring. (c) Through-port transmission under different dc bias voltages. (d) Micrograph of the transmitter with optical devices and frontend circuit features labeled. The gentle roll-off in relative optical transmission at lower wavelengths is due to the bandwidth of the grating couplers, which has a transmission peak centered at about 1220 nm.

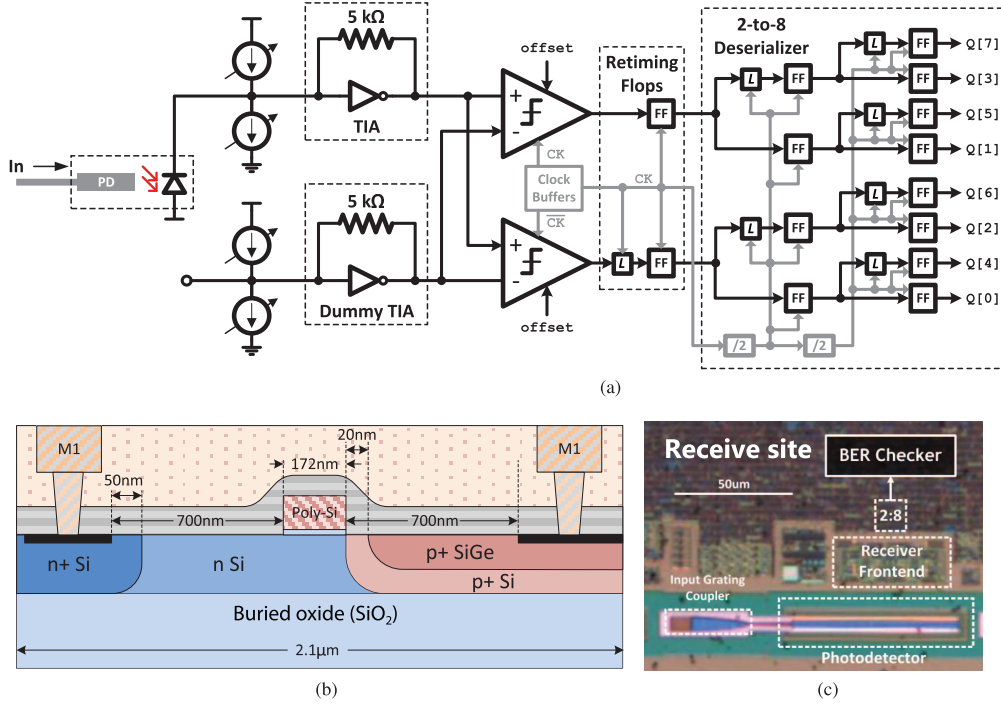


Fig. 9. (a) Schematic of the receiver frontend and deserializer. (b) Structure of the SiGe PD. (c) Micrograph of the receiver receiver site.

L_N and N_0 over 64 integration intervals in place of individual L_N and N_0 samples. The “eye-max” controller makes heating choices at a decision rate of once every 2^{15} to 2^{31} bits. Logic for heating cancellation follows (12) and produces cancellation power offsets, P_C , which is added to the heater output of the controller to form the 9 bit digital heating code P_H . The heater driver DAC converts P_H to power dissipated by the embedded microring heater.

The receiver (Fig. 9) consists of a PD and a TIA followed by two data slicers operating on even and odd clock phases. The PD is constructed using silicon-germanium (SiGe) present in the process to strain-engineer electrical p -channel transistors [36]. The low parasitic capacitance afforded by monolithic integration enables 7 GHz of bandwidth from a single-stage inverter TIA with a 5 kΩ feedback resistor. A dummy TIA provides the reference voltage to the data slicers. Capacitive and current DACs in the two data slicers (and current DACs at the TIA inputs) allow for adjustment of the receiver decision threshold, providing offset compensation and eye-monitoring. Retiming latches and flops interface the frontend circuitry to the 2-to-8 CMOS demux-tree deserializer. The bit-error-rate checker reads the deserializer output to provide *in situ* BER measurements. The PD has a responsivity of 0.023 A/W at 1180 nm and is a variant of the detector in [36] with slightly shifted geometries. The device is absorption-length-limited and could be resonated to increase the absorption [36], [37]. Standalone-resonated SiGe detectors on the platform exhibit a responsivity of 0.55 A/W [37].

V. EXPERIMENTAL DEMONSTRATION

A. Spoked-Ring Thermal Characterization

We first characterize the heater DAC and thermal properties of the spoked-ring modulator device (Fig. 10). The DAC

delivers 2 mW of heater output power, achieving 2.5 nm (524 GHz) tuning range (≈ 50 K of temperature change) and 1.25 nm/mW (3.8 μ W/GHz) DAC-included tuning efficiency. The 9-bit DAC has 1.02 GHz/LSB resolution, without any self-heating effects from the laser (which enhances or degrades the GHz/LSB). Fig. 10(c) shows the thru-port response of an ON-resonance ring ($\lambda = \lambda_0$) to a 2.5 mW (a 524 GHz change in λ_0) step applied by the heater DAC. The ring, whose $\Delta\lambda = 21.7$ GHz, crosses its 3 dB point in 300 ns, indicating that it has shifted 10.8 GHz (half of its $\Delta\lambda$) in that time, which is $1/49$ of the applied step. From this, we calculate $\tau_T \approx 14.7 \mu$ s for the spoked ring modulator.

B. Wavelength-Locking Transmitter Demonstration

We demonstrate the capabilities of the wavelength-locking transmitter in Fig. 11. During the lock-on process, the controller finds and locks to the point with maximum $|L_d|$, the eye height. A $2^{31} - 1$ PRBS sequence is used for both the lock-on training sequence and the signal for the eye-measurement. The controller decision rate is 76.3 kHz. Fig. 12 shows the states of tuner controller during lock-on, slowed 10 000 $\times$ to enable configuration I/O capture. The upper-bound on lock time is 6.7 ms, assuming a worst case P_{H-init} from the final lock position and the controller making only 1 LSB steps. Larger step sizes during *search* and *sweep* decrease the lock time. The negative feedback effects of self-heating contribute a DAC resolution enhancement of approximately $9\times$ (> 3 bits). The heater DAC steps through 90 LSB to tune the ring across half the ring's $\Delta\lambda$ (21.7/2 GHz), an effective resolution of 0.12 GHz/LSB, compared to 1.02 GHz/LSB estimated for the case without self-heating.

The transmitter achieves 5 Gb/s modulation with a 6.5 dB extinction ratio (ER) and an insertion loss (IL) of 3 dB

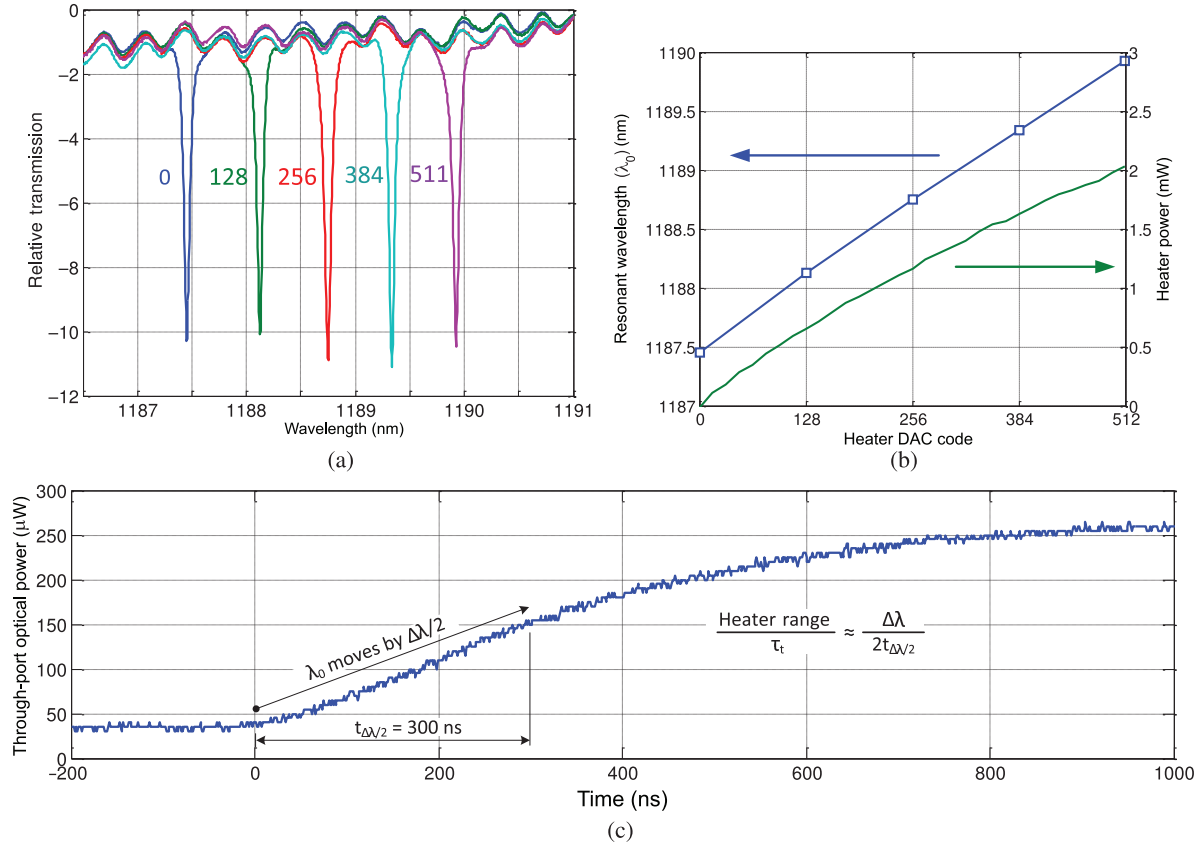


Fig. 10. (a) Characterization of the thermal characteristics of the spoked-ring modulator device. Modulator optical transmission spectrum at different heater DAC codes. (b) Spoked-ring resonant wavelength (λ_0) and heater driver power consumption versus DAC codes. (c) Transient through-port response of a ring on resonance with the laser to a heater step (DAC = 0 $\rightarrow$ 511) and calculation of τ_T .

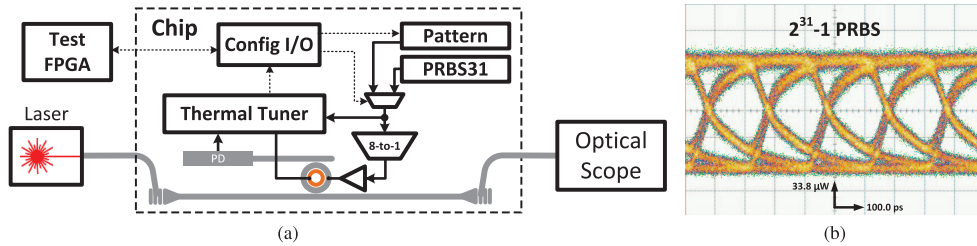


Fig. 11. (a) Test setup for the transmitter characterization and (b) 5 Gb/s $2^{31} - 1$ PRBS transmit eye diagram. The transmitter block can be configured to transmit either a $2^{31} - 1$ PRBS sequence or a 64-bit pattern. We can also monitor the status of the thermal tuner through the configuration I/O module.

[Fig. 11(b)]. We use a swing of -0.7 to 0.5 V, accomplished by setting $V_{DD} = 1.2$ V for the final drive inverters and tying the modulator anode to 0.5 V. The final drive inverters and 8-to-1 serializer consume 30 and 140 fJ/bit, respectively. We note that an implant mask error is limiting the achievable data rate of the modulator; the contacts to the p -type junctions were doped with a mid-level p -implant as opposed to a $p+$ implant, creating high contact resistance and degrading modulator transition times. Use of the correct mask will remove this speed limitation. At 5 Gb/s, the tuner frontend (integrator, SAR ADC) and backend consume 0.20 and 0.52 mW, respectively. The tuner frontend sensitivity is < 500 nA, though the drop-port PD provides ≈ 3.5 μ A of current at the laser power used.

To demonstrate the full capability of the level-tracking circuitry and the self-heating canceller, we change the ratio of

zeros (r_0) in the transmitted data. Patterns are randomly generated every 25 ms with different r_0 , including patterns with $r_0 = 1$ (all zeros) and $r_0 = 0$ (all ones). With the tuner operating in average power lock mode (accomplished by forcing $L_d = 0$, such that $L_0 = L_1 = \text{average of } 8192 \text{ bits}$), the modulator eye is completely closed as the controller is unable to discern changes in r_0 from a drift in resonance [Fig. 13(a)]. Enabling separate L_0 , L_1 , and L_d tracking allows the eye to settle at the correct value, though each change in r_0 causes a sudden change in laser self-heating power, creating temporary eye corruption [Fig. 13(b)]. The self-heating cancellation circuit compensates for this effect, eliminating the corruption [Fig. 13(b)].

We use the wavelength-locked transmitter components to construct a $11\text{-}\lambda$ DWDM transmitter macro in this platform,

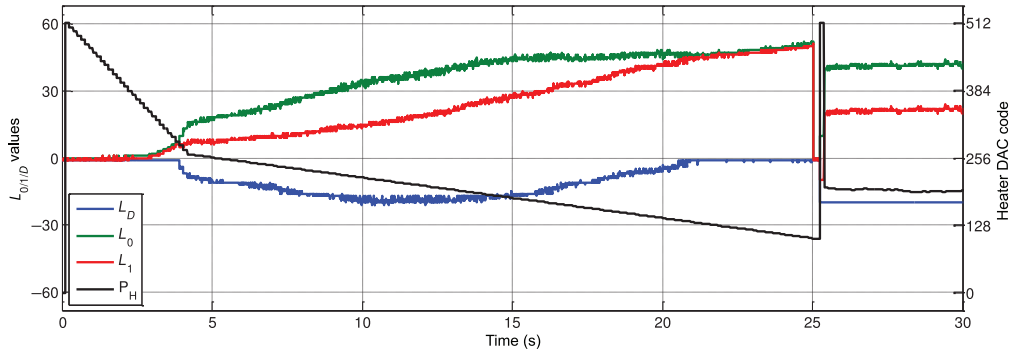


Fig. 12. Thermal tuner lock-on transient, captured experimentally, whereby a spoked-ring modulator with resting $\lambda_0 = 1187.5$ nm locks on to a laser at 1188.8 nm, slowed 10 000 $\times$. At normal speed, the controller would complete this lock in ≈ 2.5 ms. The controller makes 8 LSB strides size during *search* and 1 LSB steps during *sweep* and *lock*.

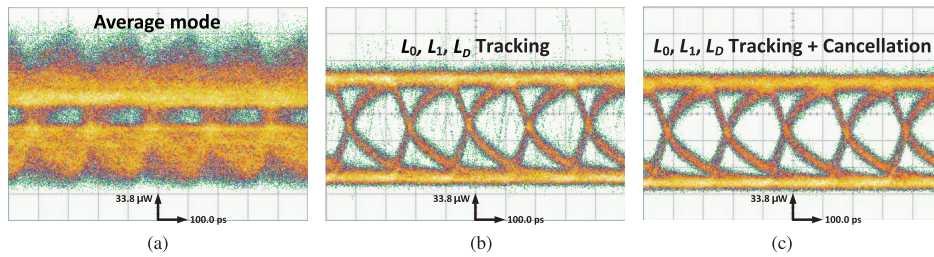


Fig. 13. (a) 5 Gb/s eye diagrams for the auto-locking transmitter when transmitting data with arbitrarily changing ratios of *ones* in the data with the tuning controller in average photocurrent tracking mode. (b) L_0, L_1, L_D tracking mode. (c) L_0, L_1, L_D tracking with self-heating cancellation mode.

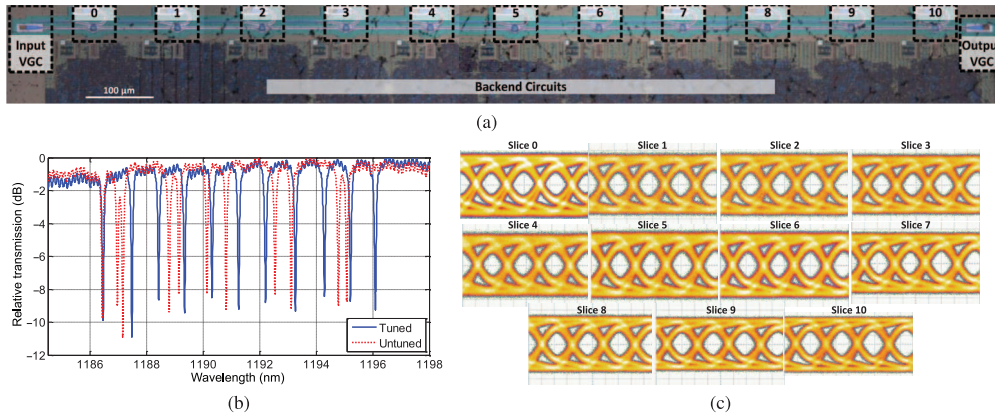


Fig. 14. (a) DWDM transmitter, (b) optical transmission spectrum before and after manual alignment to a 1.0 nm wavelength grid, and (c) 8 Gb/s eye-diagrams for each of the 11-channels, with extinction ratio of ≈ 2 dB [43].

shown in Fig. 14. Each modulator microring couples to the bus waveguide which spans across the entire macro. The radii of the microrings are stepped in increments of 10 nm, such that each ring is resonant at a different wavelength with 1.0 nm nominal channel spacing. Using the auto-locking circuitry, we demonstrate the functionality of each of the transmitter sites, one at a time, achieving 8 Gb/s open transmit eyes on each of the 11 channels. The DWDM transmitter bank can achieve up to 11×8 Gb/s aggregate transmit data rate, potentially enabling 88 Gb/s of throughput on a single on-chip waveguide or optical fiber. Given a 3.07 THz FSR, the macro achieves a spectral efficiency of 0.018 bits/s/Hz. The full macro, which includes the digital backend, thermal tuning logic, transceiver frontends, and optical devices, is $1500 \mu\text{m} \times 150 \mu\text{m}$, corresponding to a

bandwidth density of 391 Gb/s/mm². Microrings are placed at a pitch of 128 μm but can be as tight as approximately 20 μm before violating minimum metal density rules.

C. Data Receiver and Chip-to-Chip Link

We characterize the optical data receiver using the 0.023 A/W SiGe PD (Fig. 15). We use a 10 Gb/s external reference modulator to externally modulate a laser with a $2^{31} - 1$ PRBS and couple the modulated light with the chip waveguide and PD. We measure the average photocurrent flowing through the PD bias port to obtain the receiver photocurrent sensitivity. The receiver achieves $< 10^{-12}$ bit-error-rate at 10 Gb/s with average photocurrent sensitivity of 9 μA corresponding to

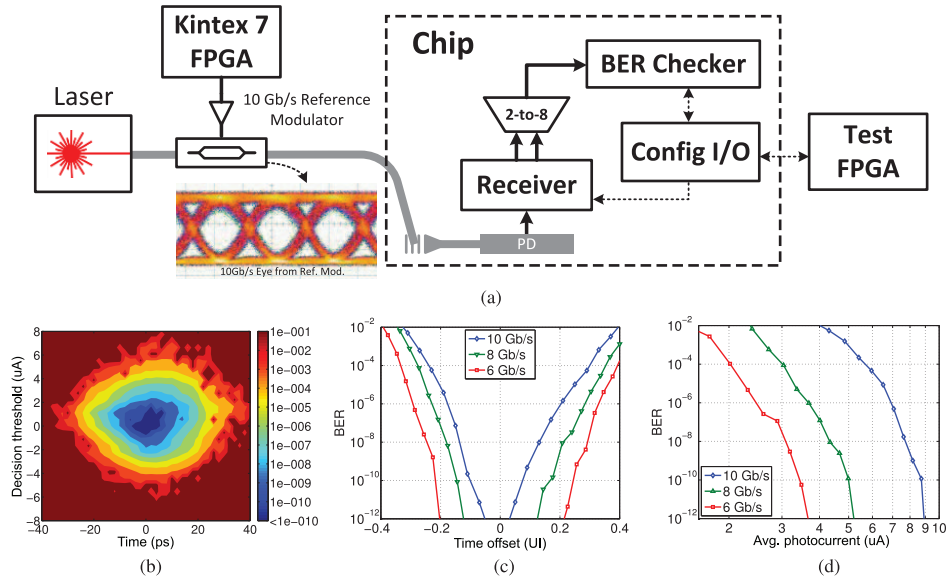


Fig. 15. (a) Receiver experimental test setup, including the 10 Gb/s eye from the external reference modulator, (b) 10 Gb/s optical BER eye diagrams, (c) receiver BER bathtub curves at different data-rates, and (d) receiver BER versus average measured photocurrent.

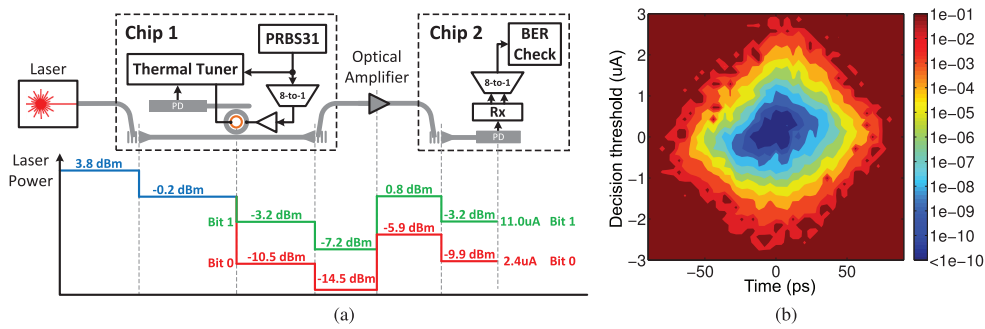


Fig. 16. (a) 5 Gb/s optical chip-to-chip link with the amount of optical power annotated at each point of the link. (b) Link BER eye diagram. The center of the eye is bit-error-free for $> 2 \times 10^{10}$ received bits.

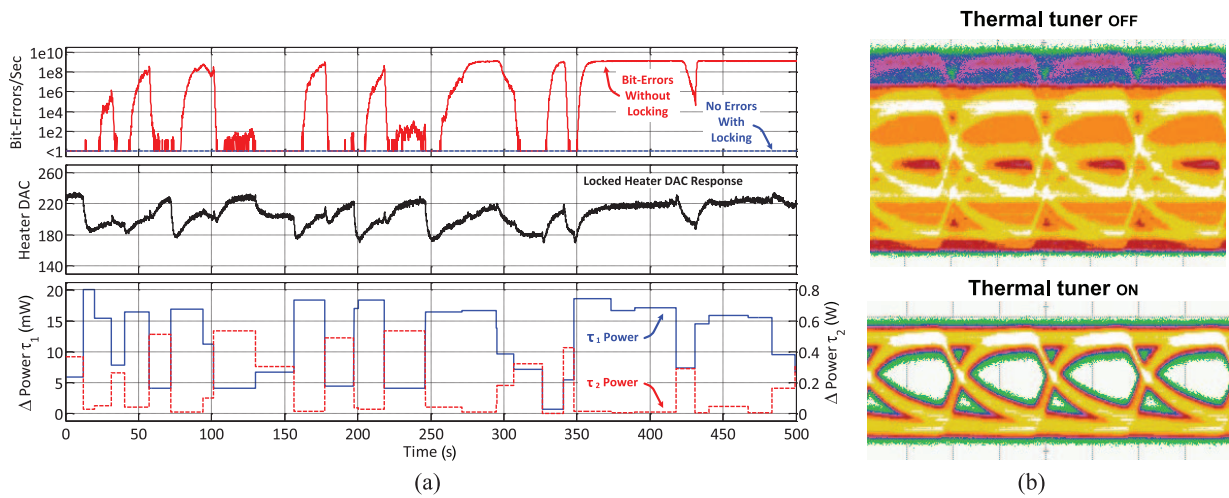


Fig. 17. (a) Link thermal stress test and (b) transmit eyes with the thermal tuner enabled and disabled over the course of this experiment. We perform this experiment at 2.5 Gb/s and estimate the maximum temperature change to be < 8 K.

−4.06 dBm optical sensitivity with a 0.023 A/W PD. Pairing the receiver with the 0.55 A/W-resonated PDs on the platform would improve optical sensitivity by 14 dB. The sensitivity degradation from 8 to 10 Gb/s [Fig. 15(d)] is partly due to the

lower quality eye from the reference modulator. At 10 Gb/s, the receiver frontend consumes 3.48 mW (348 fJ/bit), with 0.61 mW used by the TIA and 2.88 mW by the slicers and clock buffers. The 2-to-8 deserializer consumes 1.5 mW.

TABLE I
SUMMARY AND COMPARISONS

	This Work	[39]	[33]	[27, 28]	[8]	[4]
Technology						
Photonics	45 nm CMOS SOI	130 nm SOI	130 nm SOI	130 nm SOI	130 nm SOI	180 nm Bulk CMOS
Circuits		40 nm CMOS	65 nm CMOS	65 nm CMOS	40 nm CMOS	
Integration	Monolithic	Flip-chip	Wirebond	Wirebond	Flip-chip	Monolithic
Wavelength Band (nm)	1180	1550	1310	1550	1550	1280
Transmitter						
Data rate (Gb/s)	5	10	5	25	20	5
Ring radius (μm)	5	12	2.5	7.5	7.5	7.25
Ring mode	Depletion	Depletion	Injection	Depletion	Depletion	Depletion
Ring Q	$\approx 11\,600$	$\approx 15\,000$	≈ 8000	≈ 5000	≈ 5500	≈ 8000
WDM channels	11	8	1	5	4	9
Extinction (dB)	> 6	7	12.7	7	> 7	7.6
Insertion loss (dB)	3–4	5	0.9	N/R	N/R	1.8
Driver power	0.15 mW 0.03 pJ/bit	1.35 mW 0.14 pJ/bit	4.04 mW 0.81 pJ/bit	61.8 mW 2.47 pJ/bit	26.1 mW 1.31 pJ/bit	1.75 mW 0.35 pJ/bit
Serializer power	0.70 mW 0.14 pJ/bit	N/R	N/R	31.2 mW 1.25 pJ/bit	N/R	N/R
Receiver						
Data rate (Gb/s)	10	10	8	24	20	5
PD responsivity (A/W)	0.023	0.7	1.0	0.45	0.6	0.2
PD + input cap (fF)	15–20	40–60	200	90/110	≈ 80	≈ 30
Sensitivity	–4.1 dBm@10 Gb/s –6.5 dBm@8 Gb/s	–15 dBm@10 Gb/s	–9 dBm@8 Gb/s	–7.8 dBm@24 Gb/s	–7.2 dBm@20 Gb/s	–7.5 dBm@5 Gb/s –15.2 dBm@3 Gb/s
Photocurrent sens.	9.0 μA @10 Gb/s 5.3 μA @8 Gb/s	22.4 μA @10 Gb/s	126 μA @8 Gb/s	74.1 μA @24 Gb/s	115 μA @20 Gb/s	35.5 μA @5 Gb/s 6.0 μA @3 Gb/s
Power	3.48 mW 0.35 pJ/bit	3.5 mW 0.35 pJ/bit	2.2 mW 0.28 pJ/bit	17 mW [†] 0.71 pJ/bit [†]	11.6 mW 0.58 pJ/bit	8.63 mW 1.73 pJ/bit
Deserializer power	1.50 mW 0.15 pJ/bit	N/R	N/R	N/R	N/R	N/R
Tuning						
Tuning method	Closed-loop Bit-statistical	Open-loop	Closed-loop Averaging	Closed-loop Avg. / Pk-Det.	Open-loop	Closed-loop Direct-sampled
Tuner type	Thermal	Thermal	Voltage bias	Thermal	Thermal	Thermal
Tunable range	524 GHz (2.5 nm)	95/1600 GHz [‡] (0.75/12.8 nm) [‡]	49 GHz (0.28 nm)	100 GHz (0.8 nm)	N/R	90 GHz (0.5 nm)
Max heater power (mW)	2	4	N/A	5	N/R	6
Tuning efficiency	3.8 μW /GHz (1.25 nm/mW)	42/2.5 μW /GHz [‡] (0.19/3.2 nm/mW) [‡]	6.8 μW /GHz [§] (0.83 nm/mW) [§]	50 μW /GHz (0.16 nm/mW)	49 μW /GHz (0.16 nm/mW)	67 μW /GHz (0.08 nm/mW)
Control power (mW)	0.72	N/A	0.34	0.17	N/A	0.43

N/A = Not applicable, N/R = Not reported.

[†]Includes clocking power amortized over four data channels.

[‡]Measured number/estimated number for a substrate removed chip using [22].

[§]Includes the total power of the tuning control loop.

We demonstrate the transmitter, receiver, and bit-statistical thermal tuner in a chip-to-chip optical link (Fig. 16). The link operates at 5 Gb/s and uses the spoked-ring modulator, vertical grating couplers with a loss of approximately 4 dB, and the 0.023 A/W SiGe PD. In order to overcome the very limited responsivity from the connected PD and the high grating coupler loss, we insert an optical amplifier between the two chips to provide sufficient incident laser power at the PD for the receiver. The amplifier provides approximately 8 dB of optical gain, though it also degrades the extinction ratio by 0.6 dB. We note that using the 1.2 dB loss grating couplers that already

exist on the same chip [38] in the link test sites will eliminate the need for the amplifier. Use of the resonant PDs in the link would further increase the link margin and decrease the required laser power to well below a milliwatt.

We perform a link stress test of the tuner (Fig. 17) by ramping an unrelated supply voltage, creating two coupled thermal “hot spots” on the transmit chip which affect the transmit ring through different time constants ($\tau_1 < \tau_2$). With thermal tuning disabled, the transmit microring suffers unacceptable eye closure, causing many link bit-errors, while an enabled tuner allows error-free operation.

VI. COMPARISON

We summarize the performance of the platform and compare against other works in Table I. The transmitter compares favorably in energy efficiency. This is helped, in part, by the low 1.2 V_{pp} swing requirement of the modulator device. A mask revision with the proper p-contact implant should improve speed beyond 5 Gb/s. Compared to other works, the receiver circuit demonstrates a superior 9 μ A photocurrent sensitivity at 10 Gb/s, benefiting tremendously from its low input capacitance. This allows the receiver to achieve comparable optical sensitivity, despite the crippling 0.023 A/W responsivity of the connected PD. We expect the optical sensitivity to improve significantly using the 0.55 A/W-resonant PD available in the platform [37], to substitute in on the next chip revision.

Aided by the thermal isolation of the substrate removed chip, this work achieves the highest measured tuning range and tuning efficiency of any on-chip closed-loop tuner reported to date. The extra fidelity provided by the bit-statistical tuner has a modest control power overhead, which could further improve with properly set timing constraints. Interestingly, of all chip-driven tuners shown here, only this work and [39] (if substrate removed) have sufficient tuning range to cover the WDM channel spacing, which is a requirement for dynamic ring-to-channel assignment permutations for reducing tuning costs and achieving virtually unlimited temperature range tolerance [40], [41]. If not satisfied, the thermal tuner must compensate for the entire range of temperature changes, a situation where a temperature delta of just 10 K would exceed the tunable range of [4], [27], [28], [33] (and [39] if without substrate removal). These results highlight the importance of providing proper thermal isolation for silicon microring-based photonics.

VII. CONCLUSION

In order for silicon-photonics to be adopted as a next-generation interconnect technology, we must demonstrate electronic–photonic integration and robust DWDM. The ring resonator is the key device toward achieving these goals, but faces a number of challenges due to its sensitivity to process, laser power, and temperature variations. This work investigated the thermal properties of microring resonators, including self-heating. To overcome these challenges, we proposed a bit-statistical microring thermal tuner, which can handle arbitrary data sequences, enable an “eye-max” optimal controller design, and provide self-heating cancellation. The proposed tuner is a critical building block of a “zero-change” electronic–photonic platform in a commercial 45 nm CMOS-SOI process, enabling a 11 λ DWDM transmitter and a thermally-resilient chip-to-chip optical link. Together with substrate removal and other components on the platform, this work is a key enabler of robust DWDM operation in VLSI electronic–photonic systems [42].

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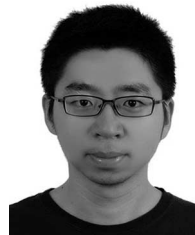
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